Lab 3: Component Modelling & Architectural Pattern Selection

Vaccination Cohort & Dose Scheduling System

Field	Details
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Domain	Healthcare & Telemedicine
Assigned Actors	Citizen Registrant, Vaccination Officer
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1. Scenario Understanding & Requirement Analysis

1.1 Problem Statement

The Vaccination Cohort & Dose Scheduling System is a public-health administration platform for organizing demographic vaccination rollouts, enforcing minimum intervals between doses, scheduling vaccination-center slots, recording administered doses, and generating verifiable QR digital vaccination certificates.

1.2 Key Functional Requirements

- Citizen Registrants register with required demographic and vaccination information.
- Eligibility is checked before vaccination-slot booking.
- Minimum dose intervals are enforced before subsequent-dose scheduling.
- Eligible citizens can view and book available vaccination-center slots without double booking.
- Vaccination Officers verify citizens and record vaccine, dose number, date, and vaccination center.
- The system generates a digitally verifiable QR vaccination certificate after dose recording.
- Digital certificates can be verified.

1.3 Key Non-Functional Requirements & Challenges

- Security & Performance: signed QR codes must be authenticated and verification must respond within 150 ms under the specified simulated peak load.
- Availability & Reliability: vaccination and appointment information must remain intact and available during campaigns.
- Booking consistency: selected slots must be revalidated so concurrent requests cannot cause double booking.

2. Architectural Style Analysis

Three candidate styles are evaluated against the vaccination system requirements: Layered, Microservices, and Client-Server Architecture.

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Style	Pros (relative to this system)	Cons (relative to this system)
Layered Architecture	Clear separation of presentation, business logic, and data; simple to understand and build.	Modules tend to scale together, making independent scaling of certificate verification or slot booking harder.
Microservices Architecture	Registration, scheduling, dose records, and certificate functions can scale/fail independently; strong service boundaries and independent deployment.	More operational complexity, network communication, and distributed data-consistency challenges.
Client-Server Architecture	Centralized control and simple deployment; straightforward data consistency.	Central server can be a bottleneck or single point of failure during campaign peaks; limited fault isolation.

3. Architecture Selection

Selected Architecture: Microservices Architecture

“We chose the Microservices Architecture for the Vaccination Cohort & Dose Scheduling System.” The system contains distinct capabilities for registration and eligibility, scheduling and dose-interval enforcement, vaccination records, and certificate generation/verification. These capabilities have different performance, security, availability, and scaling needs, making independently deployable services appropriate.

4. Component Identification

#	Component	Responsibility
1	API Gateway	Single entry point for clients; routes requests to the appropriate services.
2	Registration & Eligibility Service	Handles citizen registration and eligibility checks before scheduling.
3	Scheduling & Dose Interval Service	Handles slot search, booking, schedule updates, and minimum dose-interval enforcement.
4	Vaccination Record Service	Allows authorized officers to record administered vaccine and dose details.
5	Certificate Service	Generates certificates and authenticates digitally signed QR codes.
6	Vaccination Data Store	Reliably persists citizen, appointment, vaccination, and certificate information.

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5. UML Component Diagram

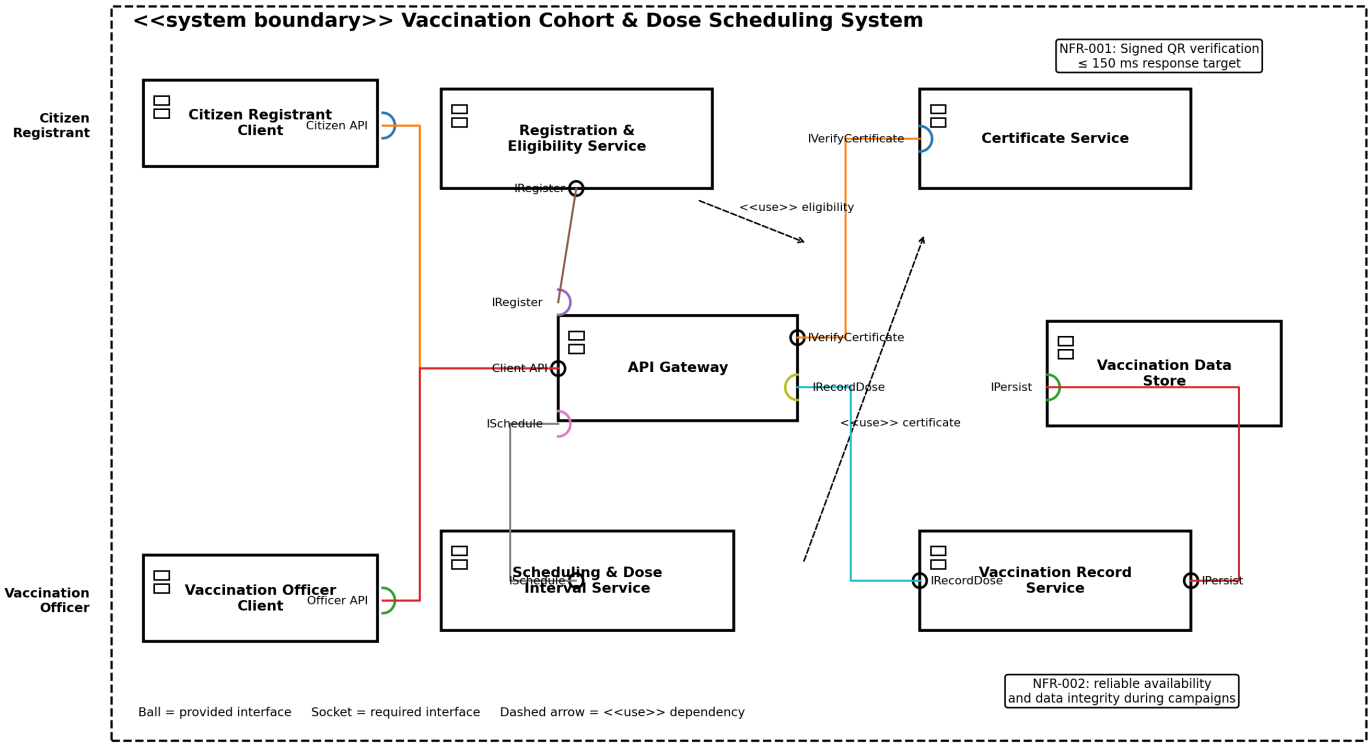


Fig. 1 — Component diagram of the Vaccination Cohort & Dose Scheduling System backend

5.1 Interface Summary

Interface	Provider → Requirer	Purpose
IRegister	Registration & Eligibility Service → API Gateway	Registration and eligibility operations for the citizen workflow.
ISchedule	Scheduling & Dose Interval Service → API Gateway	Slot search, booking, updates, and dose-interval enforcement.
IRecordDose	Vaccination Record Service → API Gateway	Authorized vaccination-dose recording operations.
IVerifyCertificate	Certificate Service → API Gateway	Certificate generation and signed QR verification; target ≤ 150 ms.
IPersist	Vaccination Data Store → Services	Persistence for citizen, appointment, vaccination, and certificate data.

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6. Written Justification

“We chose the Microservices Architecture for the Vaccination Cohort & Dose Scheduling System.”

Architectural Choice:

The system is organized as independently deployable services for registration and eligibility, scheduling and dose-interval enforcement, vaccination records, certificate generation/verification, and persistent data access, fronted by an API Gateway.

Two Scenario-Related Reasons:

1) Independent scaling for uneven load: certificate verification has a strict response-time target and can be optimized or scaled separately from registration and scheduling. 2) Fault isolation and reliability: separating major capabilities reduces the chance that a problem in one function makes unrelated vaccination workflows unavailable.

Security Advantage:

The Certificate Service can isolate the security-sensitive QR verification function and authenticate digitally signed codes before returning a result. Service boundaries also reduce unnecessary exposure of citizen and vaccination data to unrelated components.

Performance Benefit:

The certificate-verification path can be independently scaled and kept focused, supporting the specified 150 ms response target under simulated peak load. Scheduling can likewise be scaled during vaccination-campaign demand without scaling every other function.

Key Design Trade-off:

Microservices add operational complexity, network communication, and distributed data-consistency concerns. These costs are accepted because the system needs secure, low-latency certificate verification, campaign-time availability, and reliable vaccination and appointment records.